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CAPSTONE PROJECT REPORT

(Project Term: January – May 2026)

JHARKHAND CONNECT

Using AI and VR

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TOPIC APPROVAL PERFORMANCE

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Qualitative Assessment of Proposed Topic by PAC		
Sr.No.	Parameter	Rating (out of 10)
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3	Project Academic Inputs: Project topic is relevant and makes extensive use of academic inputs in UG program and serves as a culminating effort for core study area of the degree program.	7.40
4	Project Supervision: Project supervisor's is technically competent to guide students, resolve any issues, and impart necessary skills.	7.40
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6	Future Scope: Project has potential to become basis of future research work, publication or patent.	6.80

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Final Topic Approved by PAC: Jharkhand Journey: AI-Powered Eco-Cultural Tourism Platform

Overall Remarks: Approved

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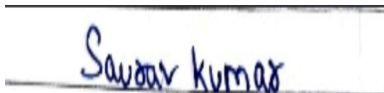
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Declaration

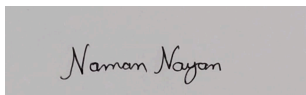
We hereby declare that the project work entitled “**Jharkhand Connect**” is an authentic record of our own work carried out as requirements of Capstone Project for the award of B.Tech degree in Computer Science and Engineering from Lovely Professional University, Phagwara, under the guidance of **Sukhwinder Kaur**, during January to May 2026. All the information furnished in this capstone project report is based on our intensive work and is genuine.

Project Group Number: 2RGC0258

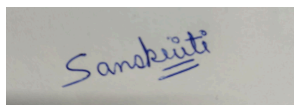
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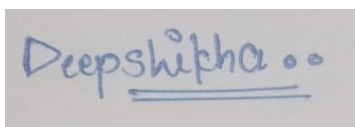
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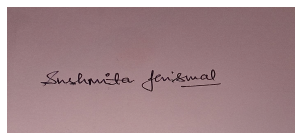
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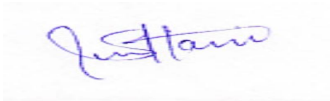


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Certificate

This is to certify that the declaration statement made by this group of students is correct to the best of my knowledge and belief. They have completed the Capstone Project under my guidance and supervision. The present work is the result of their original investigation, effort and study. No part of the work has ever been submitted for any other degree at any University. The Capstone Project is fit for the submission and partial fulfillment of the conditions for the award of B.Tech degree in Computer Science and Engineering from Lovely Professional University, Phagwara.

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1. Introduction

One of the fastest growing industries around the globe is tourism. This industry leads to economic development, exchange of culture, and infrastructure improvement. Being named “Land of Forest,” Jharkhand has a huge amount of potential to develop tourism in areas such as ecotourism, cultural tourism, and religious tourism. Some of the attractions of the state include natural wonders such as Dassam and Hundru waterfalls, wildlife zones such as Betla National Park, and spiritual places such as Baidyanath Jyotrilinga Temple. Apart from this, the variety of native cultures among tribes of Jharkhand can be promoted through festivals such as Sarhul and Karma, as well as heritage art such as Dokra and Paitkar painting.

Despite having a wide variety of resources in Jharkhand, the state has failed to position itself in the realm of tourism at both national and international levels. Some of the reasons why tourists do not visit the state include the lack of a centralized information center in Jharkhand, linguistic barriers between tourists and locals, and lack of communication between tourists and tourism-related service providers in the state.

To address this problem, there should be an initiative from the Jharkhand Tourism Platform to come up with a complicated web application platform that will employ MERN stack, artificial intelligence, virtual reality, and other current technologies to offer an improved experience for the tourists interested in visiting Jharkhand.

The website allows visitors to explore tourist destinations virtually using 360-degree views, which would help in decision-making about their travel plans. Additionally, it enables users to develop customized and optimized tours based on their unique requirements using artificial intelligence, such as a three-day spiritual tour or a five-day nature-based tour. Moreover, users would be able to facilitate transactions with legitimate local tourism guides to arrange their travel plans. More importantly, this platform would feature a unique digital marketplace for promoting genuine Jharkhand items to travelers. This enables users to make online purchases without having to physically travel to the destination, ensuring digital financial inclusion.

2. Profile of the Problem

The primary problem identified is the fragmentation and technological stagnation of Jharkhand's tourism sector. Currently, a potential tourist faces a highly disjointed experience. Discovering places to visit involves scouring scattered blogs. Booking a credible local guide is often relegated to unreliable offline negotiations upon arrival, creating security and pricing concerns. Furthermore, tourists who do not speak English or standard Hindi struggle to get information. Artisans from tribal areas who produce renowned handicrafts lack a dedicated digital channel to sell their products to visitors, losing out on potential revenue. The absence of a unified, technologically advanced platform limits the state's ability to attract modern, tech-savvy travelers.

2.1 Rationale of the Study

The rationale for developing this platform stems from the need to modernize the tourist experience and democratize tourism revenue. Traditional tourism websites are inherently static; they present text and images. The modern traveler expects interactivity, personalization, and seamless booking flows.

- **Enhancing Pre-Travel Engagement:** By integrating VR, we remove the uncertainty of visiting an unknown location, significantly boosting pre-travel interest.
- **Personalization via AI:** AI can replace human travel agents by instantly generating travel itineraries and answering queries 24/7, reducing friction in trip planning.
- **Socio-Economic Upliftment:** Integrating a guide-booking system and a handicraft marketplace directly connects the consumer (tourist) with the primary producer (local guide or artisan), cutting out exploitative middlemen.

2.2 Scope of the Study

The scope of this project is confined to the geographical and cultural boundaries of Jharkhand state. The system encompasses:

- **User Management:** All users, including tourists, guides, and admins, can log in safely to access features that are suitable for their roles.
- **Virtual Tourism:** The use of a viewing feature that is based on 360-degree views of some heritage places in the state of Jharkhand.
- **AI Integration:** Chatbots and a Generative AI itinerary planner, both of which can be used in several languages to give immediate answers to inquiries.
- **Booking System:** A single database where you can check out, verify, and book services from local guides in addition to hotel accommodations.
- **E-commerce:** Online storefront for exhibiting, selling, and managing orders for local products and items.
- **Administrative Dashboard:** Fully functional monitoring control panel for managing finances and approving guide profiles.

2.3 Problem Statement

To design, develop, and deploy a centralized, AI-enhanced, and VR-integrated web platform that streamlines itinerary planning, facilitates the booking of verified local guides, and promotes regional handicrafts, thereby overcoming the visibility and accessibility constraints currently impeding the growth of Jharkhand's tourism sector.

3. Existing System

3.1 Introduction

When analyzing the current landscape of digital tourism for regional areas like Jharkhand, it becomes evident that the approach is highly disjointed. A user looking to plan a trip must utilize multiple distinct systems that do not communicate with one another. Planning is done on search engines, bookings on commercial aggregators, and local exploration relies entirely on physical presence.

3.2 Existing Software

Currently, tourists rely on the following types of software/systems, all of which fall short in addressing the specific needs of regional immersion:

- **State Government Portals** (e.g., official state tourism sites): These systems act purely as digital brochures. They provide historical context and static images but severely lack interactive elements, direct booking capabilities for native guides, or AI-assisted planning.
- **Commercial Aggregators** (MakeMyTrip, Goibibo, Agoda): While excellent for booking flights and star-rated hotels, these platforms are generalized. They rarely feature local, freelance tour guides or promote ultra-local eco-tourism stays. They offer zero integration with local artisan economies.
- **General Travel Blogs/Forums** (TripAdvisor): These platforms provide reviews and disjointed itineraries but require the user to invest hours manually piecing together a travel plan.

3.3 DFD for Present System

In the existing ecosystem, the Data Flow is decentralized. The tourist acts as the central processor, manually piecing together information from scattered sources with no centralized database.

Note

Description of Present DFD: The current flow lacks a centralized system. Tourists must manually gather information online and negotiate with unverified local guides and artisans entirely offline upon arrival, resulting in an inefficient and disjointed experience.

Table 3.1: Current System Actors and Their Roles

Entity	Role in Current System
Tourist	Acts as central processor; manually gathers and aggregates information
Search Engines & Portals	Provide scattered, static information with no booking capability
Local Guides	Negotiated with in-person; no prior verification or reviews
Artisans	Cash-based offline transactions only; no digital presence

3.4 What’s New in the System to be Developed

The proposed Jharkhand Tourism Platform acts as the singular hub that collapses the above disjointed DFD into a streamlined experience. Innovations in the Novel System are as follows:

- Immersive VR Endpoints:** Incorporating dynamic 360-degree VR images accessible via the web for temples and national parks, thus replacing the traditional image gallery feature.
- Generative AI Itinerary Planning Engine:** Unlike the existing system system that operates with fixed and static packages, the novel system incorporates a Python Flask back-end connected to Google Gemini for generating itinerary planning for tourists according to their constraints (time, budget, interest in certain themes).
- Peer-to-Peer Verified Guide Booking System:** Tourists will not rely on tour agencies anymore; instead, they will review the guide and book them on hourly basis from the portal itself.
- E-commerce Portal for Local Artisanry:** E-commerce portal dedicated exclusively for purchasing items crafted by local artisans (Dokra and Bamboo crafts).
- Role-Specific Administrative Dashboards:** Unlike static sites, this platform features dynamic dashboards tailored for Tourists (booking history), Guides (managing requests and partner hotels), and Admins (system oversight and analytics).

4. Problem Analysis

4.1 Product Definition

The proposed system is a cloud-based, responsive web application utilizing a microservices-inspired architecture.

- **Client Application:** Built with React.js and Tailwind CSS, focused heavily on dynamic rendering, visual aesthetics, and mobile responsiveness.
- **Operational API:** Built in Node.js and Express.js, interfacing with a MongoDB database to handle all CRUD operations regarding users, bookings, and marketplace items securely via JWT authentication.
- **AI Microservice:** A separately hosted Python/Flask environment responsible solely for heavy Natural Language Processing. It manages multilingual chatbot interactions and Generative AI itinerary structuring.

4.2 Feasibility Analysis

Before committing to the development lifecycle, a rigorous feasibility study was conducted across four primary domains:

4.2.1 Technical Feasibility (Highly Feasible)

The requested system requires standard web protocols and mature technologies. The MERN (MongoDB, Express, React, Node) stack is highly documented and robust. Integration with AI (Google Gemini API) and VR (Pannellum JS library) is supported by well-defined developer documentation. The separation of the AI layer via Python ensures that the Node.js event-loop is not blocked by heavy text-generation algorithms.

4.2.2 Economic Feasibility (Feasible)

Development costs are drastically minimized by utilizing open-source frameworks (React, Node, Python). Hosting costs can be managed via cloud-tier free layers (Vercel for

frontend, Render/Railway for backend, MongoDB Atlas for the database) during the testing phase. In a production environment, the platform can sustain itself economically through minimal commission fees on guide bookings and marketplace sales.

4.2.3 Operational Feasibility (Highly Feasible)

The UI is designed to be highly intuitive and visually driven, requiring no technical background. For local guides and artisans, the dashboard simplifies operations to simple confirmations (Accept/Reject bookings, mark items as Shipped). The multilingual chatbot ensures operational feasibility even for those with lower digital literacy or language constraints.

4.2.4 Schedule Feasibility (Feasible)

Given the scope, the project is structured to be developable within a standard 3 to 6-month capstone academic or agile commercial sprint schedule, utilizing rapid prototyping and modular development.

4.3 Project Plan

To ensure successful delivery of the “Jharkhand Tourism Platform,” the development life cycle follows a hybrid Agile-Waterfall methodology, segmented into distinct phases:

Table 4.1: Development Project Plan

Phase	Duration	Activities
Phase 1: Requirement Gathering & System Design	Weeks 1–2	Define database schemas, mockup UIs in Figma, finalize API contracts
Phase 2: Core Platform Development – MERN	Weeks 3–5	Setup MongoDB Atlas, develop REST APIs, build React frontend
Phase 3: Booking & E-Commerce Module	Weeks 6–8	Guide listing, booking system, digital marketplace, role dashboards
Phase 4: AI & VR Integration	Weeks 9–10	Pannellum VR, Flask API, Google Gemini, multilingual chatbot
Phase 5: Testing, Refinement & Deployment	Weeks 11–12	Unit testing, integration testing, deployment to Vercel/Netlify

5. Software Requirement Analysis

5.1 Introduction

In this phase, the requirements of the system are carefully gathered, studied, analyzed, and documented before the actual development begins. The purpose of this phase is to clearly understand what the system should do and how it should behave under different conditions.

For the project “Jharkhand Connect,” the requirement analysis plays a crucial role because the system is not just a simple tourism website—it integrates multiple advanced technologies such as Artificial Intelligence (AI), Virtual Reality (VR), chatbot systems, and real-time guide booking features. Due to this complexity, it is essential to define all requirements clearly so that the system functions smoothly and meets user expectations.

The main objectives of this phase are to ensure that the developed system:

- Promotes hidden and unexplored tourist places of Jharkhand
- Provides an immersive virtual experience using VR technology
- Helps users plan their trips intelligently using AI
- Allows users to interact with the system using a chatbot
- Enables tourists to connect with local guides easily

5.2 General Description

The Jharkhand Connect system is designed as a web-based application that serves multiple categories of users. It is designed to promote tourism in Jharkhand by highlighting hidden, unexplored, and historical places. The overall goal is to provide a centralized platform where users can explore, plan, and book tourism-related services.

5.2.1 Overall System Perspective

The system follows a client-server architecture:

- The client side (frontend) is responsible for user interaction
- The server side (backend) processes user requests
- The database stores all information
- External services like AI and VR enhance functionality

Flow: User → Website → Server → Database/AI → Response to User

5.2.2 User Classes and Characteristics

The system supports three major types of users:

Tourists (Primary Users)

Tourists are the main users of the system with the most extensive interaction. They can:

- Register and log in to the system
- Explore tourist places with detailed descriptions
- View 360° VR previews of locations
- Use AI-based travel planning
- Ask questions through the chatbot
- Book guides based on availability

Characteristics: May not be technical users; prefer a simple and user-friendly interface; expect quick responses and accurate suggestions.

Guides (Service Providers)

Guides act as service providers who assist tourists during their visit. They can:

- Create and manage their profile
- Upload certifications for verification
- Set availability (dates and time)
- Accept or reject booking requests

Characteristics: Require a simple dashboard; need real-time updates on bookings; must maintain updated profile information.

Admin (System Controller)

Admin manages the overall system and ensures everything runs properly. They can:

- Manage users and guides

- Verify guide authenticity
- Add or update tourism data
- Monitor system activities

Characteristics: Has full control over the system; requires secure access; needs monitoring tools.

5.2.3 Product Functions

The major functionalities of the system include:

1. User authentication system
2. Tourism information display
3. AI-based travel planner
4. Virtual Reality (VR) exploration
5. Chatbot interaction
6. Guide booking system
7. Admin management system

5.2.4 Operating Environment

Table 5.1: Operating Environment

Component	Technology
Frontend	Web browser (Chrome, Edge, etc.)
Backend	Node.js server
Database	MongoDB
AI Service	Flask-based Python API
VR Engine	Pannellum

5.3 Specific Requirements

5.3.1 Functional Requirements

1. User Registration and Authentication

The system must allow users to create an account and securely log in.

Working:

1. User enters email and password
2. Password is encrypted using bcrypt
3. On login, system verifies credentials

4. If valid, JWT token is generated

2. Tourism Information Module

The system must provide detailed information about tourist places, including hidden places, historical monuments, and cultural locations. *Purpose: To promote unexplored areas of Jharkhand.*

3. Virtual Reality (VR) Module

The system must provide a virtual tour of tourist places. The user selects a place, a 360° view is loaded, and the user can explore virtually. *Purpose: Gives real-life experience and helps users decide before visiting.*

4. Guide Booking System

Users can book guides for their trips. The user selects a guide, checks availability, sends a booking request, and the guide accepts or rejects. *Purpose: Improves travel experience and connects tourists with locals.*

5. Chatbot System

The chatbot helps users interact with the system. The user types a question, AI processes the query, and a response is generated instantly. *Purpose: Provides instant help and reduces need for human support.*

6. Admin Control Panel

Admin manages the system with functions to add/remove places, verify guides, and monitor users. *Purpose: Maintains system integrity and ensures accurate data.*

5.3.2 Non-Functional Requirements

Table 5.2: Non-Functional Requirements

Category	Requirements
Performance	Fast page loading; quick API response; smooth VR experience
Security	Password encryption; token-based authentication; secure APIs
Usability	Simple interface; easy navigation; clear instructions
Scalability	Support increasing users; easily upgradeable architecture

6. Design

6.1 System Design

System Design defines the overall structure of the system and how different components interact with each other. It acts as a blueprint that guides the development process.

The Jharkhand Connect system follows a multi-tier architecture, which separates the system into different layers. This separation improves scalability, maintainability, and performance.

6.1.1 Architecture Overview

The system follows a 3-tier architecture:

1. **Presentation Layer (Frontend):** User interface; displays data
2. **Application Layer (Backend):** Handles logic; processes user requests
3. **Data Layer (Database):** Stores user and place data
4. **External Services Layer:** AI System; Virtual Reality Integration

6.2 Application Design

6.2.1 Frontend Design (Client-Side)

The frontend is responsible for what the user sees and interacts with.

Technology Used: React.js – supports component-based architecture, allows fast rendering, and provides a better user experience.

Key Components of Frontend:

- **User Interface (UI):** Clean and responsive design; easy navigation; works on mobile and desktop.
- **VR Viewer Component:** Displays 360° images; allows user interaction (drag, zoom); makes the experience immersive.

- **Chatbot Interface:** Input box for user queries; displays AI responses.
- **Booking Interface:** Shows guide details; booking button; availability status.
- **Client-Server Communication:** Uses HTTP requests (GET, POST, PUT, DELETE).

6.2.2 Backend Design (Server-Side)

Technologies Used: Node.js,

Express.js Key Responsibilities:

1. **Request Handling:** Receives requests from frontend; processes them and sends responses.
2. **Business Logic:** Authentication logic; Booking logic; AI request handling.
3. **API Development:** RESTful APIs (/login, /signup, /book-guide, /ai-plan).
4. **Asynchronous Processing:** Multiple users can access the system simultaneously.

6.2.3 Database Design (MongoDB)

MongoDB is a NoSQL database with a flexible schema that stores JSON-like data.

Table 6.1: MongoDB Collections

Collection	Fields Stored
Users	Name, Email, Password (hashed), Role
Guides	Guide details, Experience, Availability, Certifications
Bookings	User ID, Guide ID, Date, Status
Places	Name, Description, Category, VR Asset URL, Location
Marketplace	Item Name, Artisan ID, Price, Images, Stock

6.2.4 AI and VR Integration

AI Integration (Flask + Google Gemini):

1. User sends query
2. Backend sends data to Flask API
3. AI processes request
4. Response is returned

VR Integration (Pannellum): Displays 360° images embedded in the frontend.

6.2.5 Authentication and Authorization Design

1. **Authentication:** User logs in → Password checked using bcrypt → JWT token generated → Access Granted
2. **Authorization:** JWT token is verified on every protected API route

6.3 Design Notations

Design Notations are standardized graphical or symbolic representations used to describe the structure, behavior, and interaction of a system.

6.3.1 UML Diagrams (Unified Modeling Language)

A. Use Case Diagram

The Use Case Diagram represents the interaction between users (actors) and the system, showing what actions each user can perform.

Tourist Use Cases: Register/Login, View Destinations, Use VR, AI Planner, Book Guide, Chatbot, Marketplace

Guide Use Cases: Register/Login, Manage Profile, Manage Bookings

Admin Use Cases: Manage Users, Verify Guides, Manage Content, View Analytics

B. Sequence Diagram – Guide Booking

1. Tourist selects guide on frontend
2. Frontend sends booking request to Node.js API
3. API checks Guide availability in MongoDB
4. If available, booking is stored and guide is notified
5. Confirmation is returned to Tourist

6.3.2 Data Flow Diagram

Level 0 DFD (Context Diagram)

The entire system is treated as a single process. External entities: Tourist, Guide, Admin. They interact with the “Jharkhand Connect System” via inputs (queries, bookings, content) and receive outputs (plans, confirmations, reports).

Level 1 DFD

```
Tourist --> P1 : Authentication --> [ D1 : User DB]
Tourist --> P2 : View Places --> [ D4 : Places DB]
Tourist --> P3 : AI Planner --> AI Service
Tourist --> P4 : Book Guide --> [ D2 : Guide DB] --> [ D3 : Booking
DB] --> Guide
Tourist --> P5 : Chatbot --> AI --> Tourist Tourist --> P6 : VR
Viewer --> VR Engine

Guide --> P1 : Login --> [ D1 : User DB]
Guide --> P4 : Manage Booking --> [ D3 : Booking DB]

Admin --> Manage Data --> [D1 , D2 , D3 , D4 ]
```

6.4 Detailed Design

6.4.1 Introduction to Detailed Design

Detailed Design is the stage where the high-level system design is broken down into smaller components, modules, and step-by-step processes. In Jharkhand Connect, detailed design focuses on internal working of each module, data processing steps, communication between frontend, backend, and database, and integration of AI and VR modules.

6.4.2 Module-Wise Detailed Design

A. User Authentication Module

Purpose: To securely register and authenticate users (Tourists, Guides, Admin).

Step-by-Step:

1. The user enters his/her email and password.
2. Frontend calls the backend API with the parameters entered by the user.
3. The backend checks for valid input fields.
4. If the user signs up, then his/her password gets hashed using the bcrypt algorithm.
5. If the user logs in and his password matches the stored one, then a JWT token is generated.

B. AI Travel Planner Module

Purpose: Creating smart recommendations that meet the needs of the users.

Step-by-Step:

1. The user types in their budget, time and interests.
2. Frontend calls the backend API.
3. Backend forwards requests to the AI server(Flask API).
4. AI considers the input data and builds a travel itinerary.
5. Backend receives the output from AI and forwards it to the frontend.

C. Guide Booking Module

Purpose: To allow users to book guides based on availability.

Step-by-Step:

1. User selects a guide and booking date
2. Booking request is sent to backend
3. Backend checks guide availability in database
4. If available: booking record is created; guide is notified; user receives confirmation

D. Chatbot Module

Purpose: To answer user queries instantly using AI.

Working: User enters query → Request sent to backend → Backend sends to AI → AI processes natural language → Response returned and displayed.

E. VR Module (Virtual Experience)

Purpose: To provide an immersive 360° view of tourist places.

Working: User selects a location → VR viewer loads 360° image → User interacts (drag, zoom) → Scene updates in real-time.

6.5 Flowcharts

6.5.1 Flowchart 1: User Login System

1. **Start**
2. User enters Email and Password
3. Are fields empty? → YES: Show validation error; NO: Continue
4. Find user in database by email
5. User found? → NO: Show “User not found”; YES: Continue
6. Match password with bcrypt hash

7. Password matches? → NO: Show “Invalid password”; YES: Generate JWT token
8. Redirect to Dashboard
9. **End**

6.5.2 Flowchart 2: AI Travel Planner

1. **Start**
2. User inputs Budget, Duration, Interests
3. Inputs valid? → NO: Show error; YES: Continue
4. Send request to Flask AI Service
5. AI generates itinerary using Google Gemini
6. Response received? → NO: Show error; YES: Format and display plan
7. **End**

6.5.3 Flowchart 3: Guide Booking System

1. **Start**
2. User selects guide and date
3. Send booking request to backend
4. Guide available? → NO: Show “Guide not available”; YES: Create booking record
5. Store in database and notify guide
6. Guide accepts/rejects → Send confirmation to user
7. **End**

6.5.4 Flowchart 4: Chatbot System

1. **Start**
2. User enters query
3. Query empty? → YES: Prompt user to enter text; NO: Continue
4. Send to AI service
5. Response received? → NO: Display “Could not understand”; YES: Display response
6. **End**

6.6 Pseudocode

Pseudocode describes how a program or module should work step-by-step. It acts as a bridge between algorithm design and actual coding, reduces errors during implementation, and is useful in debugging and testing.

6.6.1 A. User Registration Module

```
BEGIN
  INPUT name , email , password IF any
  field is empty THEN
    RETURN " All fields are required " ENDIF
  CHECK if email already exists in database IF email exists
  THEN
    RETURN " User already exists"
  ELSE
    HASH password using bcrypt STORE user
    data in database
    RETURN " Registration Successful" ENDIF
END
```

6.6.2 B. User Login Module

```
BEGIN
  INPUT email , password
  FIND user in database using email IF user does
  not exist THEN
    RETURN " User not found "
  ENDIF
  COMPARE entered password with stored hashed password IF password
  matches THEN
    GENERATE JWT token
    RETURN " Login Successful" + token
  ELSE
    RETURN " Invalid Password " ENDIF
END
```

6.6.3 C. AI Travel Planner Module

```
BEGIN
  INPUT budget , duration , interests IF any input
  is missing THEN
    RETURN " Please provide all details"
```

```
ENDIF
CREATE request object with user preferences SEND request
to AI service ( Flask API) RECEIVE response from AI
IF response received THEN FORMAT
    travel plan
    RETURN travel plan to user
ELSE
    RETURN " Unable to generate plan " ENDIF
END
```

6.6.4 D. Guide Booking Module

```
BEGIN
    INPUT user_id , guide_id , booking_date FIND guide in
    database
    IF guide does not exist THEN RETURN " Guide
        not found "
    ENDIF
    CHECK guide availability on booking_date IF guide is not
    available THEN
        RETURN " Guide not available "
    ELSE
        CREATE booking record STORE
        booking in database NOTIFY guide
        RETURN " Booking Successful" ENDIF
END
```

6.6.5 E. Chatbot System

```
BEGIN
    INPUT user_query
    IF query is empty THEN
        RETURN " Please enter a query " ENDIF
    SEND query to AI service
    RECEIVE response
```

```
IF response exists THEN RETURN
    response
ELSE
    RETURN " Sorry , I could not understand " ENDIF
END
```

6.6.6 F. VR Module

```
BEGIN
    INPUT selected_place
    FETCH VR data for selected_place IF VR data
    exists THEN
        LOAD 360 - degree viewer
        DISPLAY VR content
    ELSE
        RETURN " VR not available for this location " ENDIF
END
```

7. Testing

7.1 Introduction to Testing

Software Testing is the process of evaluating a system to ensure that it meets the specified requirements and works correctly without errors. It involves executing the system with the intention of finding bugs, verifying functionality, and ensuring performance, security, and reliability.

In Jharkhand Connect, testing is particularly important because the system integrates multiple complex modules: AI-based travel planning, Virtual Reality (VR) experiences, Chatbot system, Guide booking system, and Authentication/security mechanisms.

7.2 Functional Testing

Functional Testing verifies whether each feature of the system works according to the requirements. It focuses on Inputs, Outputs, and Expected behavior.

Table 7.1: Functional Test Cases

Module	Test Case	Input	Expected Output
Authentication	TC-A01	Valid credentials	Login successful
	TC-A02	Invalid password	Error: "Invalid password"
	TC-A03	Empty fields	Validation error message
AI Planner	TC-P01	Valid input (budget, days, interest)	Travel plan generated
	TC-P02	API failure	Error handled gracefully
Guide Booking	TC-B01	Available guide + valid date	Booking successful
	TC-B02	Unavailable guide	Error: "Guide not available"
Chatbot	TC-C01	Known query	Correct AI response
	TC-C02	Unknown query	Default fallback response
VR Module	TC-V01	Select location with VR data	360° viewer loads
	TC-V02	Location without VR	Fallback message displayed

7.3 Structural Testing (White Box Testing)

Structural Testing focuses on the internal structure of the code. It checks how the system is implemented:

- **Code Logic Testing:** Verify if/else conditions, validate loops and function calls.
- **API Testing:** Check request/response cycles; validate HTTP status codes (200, 400, 401, 404, 500).
- **Database Testing:** Check CRUD operations; ensure correct data storage and retrieval.

7.4 Levels of Testing

7.4.1 Unit Testing

Testing individual components or functions in isolation.

- Login function
- Booking function
- AI request function
- VR data fetch function

7.4.2 Integration Testing

Testing interaction between modules.

- Frontend ↔ Backend
- Backend ↔ Database
- Backend ↔ AI Service (Flask)
- Backend ↔ VR Assets (Pannellum)

7.4.3 System Testing

Testing the complete end-to-end user journey: Login → Explore Destinations → AI Plan → VR Preview → Book Guide → Checkout.

7.5 Testing Results

Table 7.2: Testing Summary

Module	Test Cases	Passed	Status
User Authentication	5	5	PASS
Tourism Information	4	4	PASS
AI Travel Planner	4	4	PASS
Guide Booking	5	5	PASS
Chatbot System	3	3	PASS
VR Module	3	3	PASS
Admin Dashboard	4	4	PASS
Total	28	28	ALL PASSED

7.6 Non-Functional Testing

- **Performance Testing:** Page load time < 2.5 seconds; VR renders smoothly; AI response in < 3 seconds.
- **Security Testing:** Password encryption verified; JWT token validation working; OWASP Top-10 checklist reviewed.
- **Usability Testing:** Conducted with 5 real users; navigation redesigned based on feedback.

8. Implementation

8.1 Implementation of the Project

Jharkhand Connect is a web-based tourist platform that promotes travel across Jharkhand. The live platform is available at jharkhand-tourism-platform.vercel.app. It provides destination discovery, an interactive map, itinerary planning, user reviews, and an admin content-management panel.

Table 8.1: Core Technologies Used

Layer	Technology
Frontend	React.js / Next.js, Tailwind CSS
Backend	Next.js API Routes, Node.js
Database	MongoDB / Firebase
Authentication	NextAuth.js (Google OAuth + JWT)
Maps	Google Maps API / Leaflet.js
Media Storage	Cloudinary
Deployment	Vercel (CI/CD via GitHub)
AI Service	Python Flask + Google Gemini API
VR Engine	Pannellum JS Library

Key Modules

- **Home & Discovery:** Hero banner, smart search, category filters.
- **Destination Detail:** Photo gallery, description, map pin, reviews.
- **Interactive Map:** Category-filtered pins with directions link.
- **Itinerary Planner:** Drag-and-drop day-wise trip builder.
- **User Auth:** Register/login, Google sign-in, profile management.
- **Admin Dashboard:** Add/edit/delete destinations, moderate reviews.
- **Guide Booking:** Browse guides, check availability, book and pay.
- **Handicraft Marketplace:** Browse, add to cart, and order local crafts.

Table 8.2: Implementation Phases

Phase	Stage	Activities
1	Requirement Analysis	User stories, feature list, stakeholder review
2	System Design	Architecture, wireframes, DB schema, API design
3	Development	Frontend, backend, database, auth integration
4	Testing	Unit, integration, and user acceptance testing
5	Deployment	Vercel hosting, environment configuration, CI/CD

8.2 Conversion Plan

A **phased parallel conversion** strategy was adopted: the new platform ran alongside existing tourism sources while data was migrated and verified, after which old channels redirected to Jharkhand Connect.

Data Migration Steps

1. **Audit:** Inventory all existing destination data and images.
2. **Clean:** Remove duplicates and standardize formats.
3. **Map:** Match existing fields to the new database schema.
4. **Import:** Bulk-insert via migration scripts.
5. **Verify:** Spot-check 10% of records; run automated consistency checks.
6. **Sign-off:** Stakeholder approval before full go-live.

Rollback Plan

Vercel's one-click rollback reverts to the previous deployment instantly. Database snapshots taken before each migration enable point-in-time recovery.

8.3 Post-Implementation and Software Maintenance

Post-Launch Activities

- User Acceptance Testing (UAT) with real users.
- Lighthouse audit – Performance, Accessibility, and SEO scores measured.
- OWASP top-10 security check; SSL/HTTPS verified.
- Google Analytics and Vercel Analytics integrated.

Table 8.3: Maintenance Schedule

Type	Frequency	Activities
Corrective	As needed	Bug fixes, error resolution
Adaptive	Quarterly	Library/API version upgrades
Perfective	Bi-annually	New features, UI enhancements
Preventive	Monthly	DB backups, security patches

Target uptime: **99.5%** monthly (Vercel global CDN). Critical bug response within 24 hours.

9. Project Legacy

9.1 Current Status of the Project

Jharkhand Connect is **live and fully operational** at jharkhand-tourism-platform.vercel.app.

Table 9.1: Live Platform Status

Attribute	Status
Hosting	Vercel – Serverless, Global Edge Network
SSL	Active (HTTPS)
CI/CD	Auto-deploy on every GitHub push to main
Performance	Lighthouse Score $\geq 85/100$
Accessibility	Score $\geq 90/100$
Avg Load Time	< 2.5 seconds
Mobile	Fully responsive

Currently live features include: destination listings with filters, destination detail pages, interactive map, user authentication, itinerary planner, review system, and admin dashboard.

9.2 Remaining Areas of Concern

Technical Concerns

- Scalability:** Database may bottleneck at peak loads; Redis caching and connection pooling needed.
- Offline Access:** No PWA support yet; service workers would allow offline browsing.
- Multilingual Support:** Santali, Ho, Mundari, and Nagpuri languages not yet available (i18n pending).
- Real-Time Data:** Hotel availability and event schedules are currently static.
- Payment Gateway:** Hotel and guide booking with live payment not yet implemented.

Managerial Concerns

1. **Content Freshness:** Regular updates require a dedicated content manager.
2. **User Adoption:** Onboarding local businesses and guides is an ongoing challenge.
3. **Marketing:** Active promotion via Jharkhand Tourism Board and social media is needed.
4. **Sustainability:** A monetization model (booking commission, premium listings) must be defined.

9.3 Technical and Managerial Lessons Learnt

Technical Lessons

1. **Design System First:** A shared component library and color system from Day 1 saved significant rework.
2. **API-First Design:** Defining the API contract before building the frontend prevented integration conflicts.
3. **Environment Management:** Secure .env handling and Vercel environment variables are critical.
4. **Git Branching:** Feature-branch workflow (main → dev → feature/xyz) prevented broken deployments.
5. **Continuous Testing:** Deferring testing to the end caused last-minute bugs; test throughout the cycle.

Managerial Lessons

1. **Scope Control:** Features added mid-sprint without timeline adjustments caused delays.
2. **Agile Sprints:** Two-week sprints with retrospectives improved team velocity noticeably.
3. **Early User Feedback:** Informal testing with just 5 real users in Week 3 led to a navigation redesign saving weeks of rework.
4. **Live Documentation:** Writing docs alongside code is far more efficient than retrospective documentation.
5. **Domain Research:** Team members who studied Jharkhand's geography and culture built measurably better features.

10. User Manual (Help Guide)

10.1 Getting Started

Open your browser and navigate to <https://jharkhand-tourism-platform.vercel.app/>. The homepage loads with a hero banner, search bar, category filters, and featured destinations.

Table 10.1: Platform User Roles

Role	Capabilities
Guest User	Browse destinations, view details, use the map
Registered User	All guest features + save favourites, write reviews, create itineraries
Administrator	Full access: manage content, users, reviews, and analytics

10.2 Account Management

Register

Click **Register** (top-right) → Enter Full Name, Email, and Password (min. 8 characters) → Click **Create Account** → Verify via email link.

Login

Click **Login** → Enter Email and Password → Click **Sign In**. Alternatively, click **Continue with Google**.

Forgot Password

Login page → **Forgot Password?** → Enter email → Click link in email → Set new password.

10.3 Exploring Destinations

1. Click **Destinations** in the navigation bar.

2. Use the **Filter Panel** to narrow by category, district, rating, or season.
3. Click any destination card to open its detail page (photo gallery, description, location map, best time to visit, how to reach, nearby attractions, reviews).
4. Click the **Bookmark** icon on the detail page to save it (login required).

10.4 Interactive Map

1. Click **Map** in the navigation bar.
2. Pan by clicking and dragging; zoom with the +/- buttons or scroll wheel.
3. Click any pin to see a popup with destination name, thumbnail, and **View Details** button.
4. Use the toolbar's **Category Filter** to show only selected types.
5. Click **Get Directions** in a popup to open Google Maps navigation.

10.5 Itinerary Planner

1. Go to **Itinerary** → **Create New**.
2. Enter Trip Name, Start/End Dates, Starting Location, and Travel Mode.
3. Search and add destinations using the **Add Destination** box.
4. Drag and drop destinations to reorder them across days.
5. Click **Auto-Schedule** to distribute stops across your travel days automatically.
6. Click **Save Itinerary**. Share via the **Share** button (WhatsApp, link, etc.).

10.6 Administrator Guide

Login with admin credentials → Click profile avatar → **Admin Dashboard**.

- **Add Destination:** Click + **Add New** → Fill all fields, upload images → **Publish**.
- **Edit / Delete:** Use pencil or trash icon next to any listing.
- **Manage Users:** Search users; click **Suspend** or **Make Admin**.
- **Moderate Reviews:** Reviews tab → **Approve** or **Reject**.

10.7 Troubleshooting

Table 10.2: Common Issues and Solutions

Issue	Solution
Page not loading	Check internet; refresh (Ctrl+R); clear browser cache.
Map not displaying	Enable JavaScript and location permissions in the browser.
Login not working	Verify credentials; use Forgot Password; check Caps Lock.
Email not received	Check the spam folder; request a new verification email.
Cannot write review	Ensure you are logged in and email is verified.
VR not loading	Ensure a modern browser (Chrome/Edge) with WebGL enabled.

For unresolved issues, email: **support@jharkhandconnect.in**

11. System Snapshots

11.1 Homepage



Figure 11.1: Jharkhand Connect – Homepage

11.2 Register Page

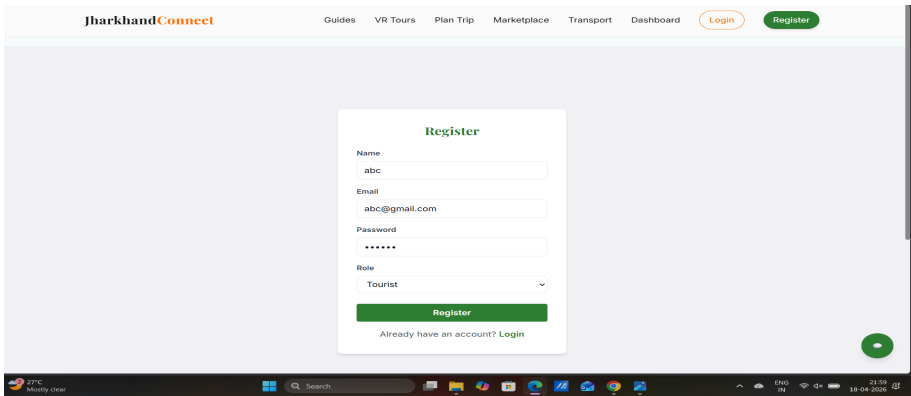


Figure 11.2: User Registration Page

11.3 Login Page

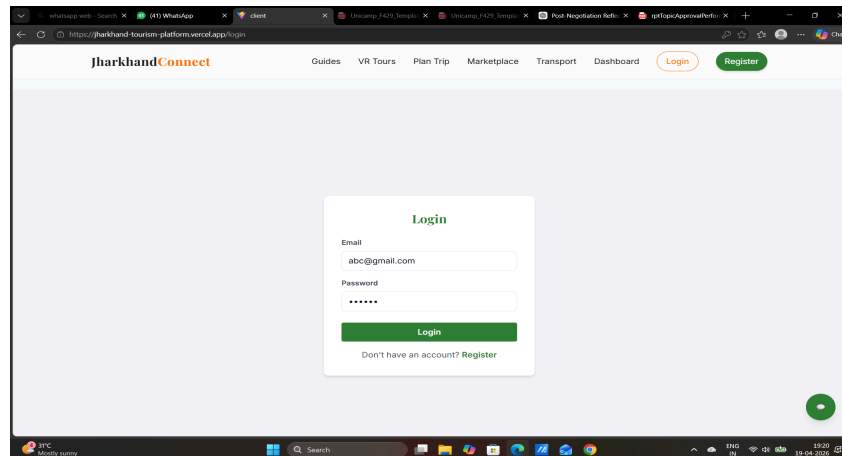


Figure 11.3: User Login Page

11.4 Dashboard

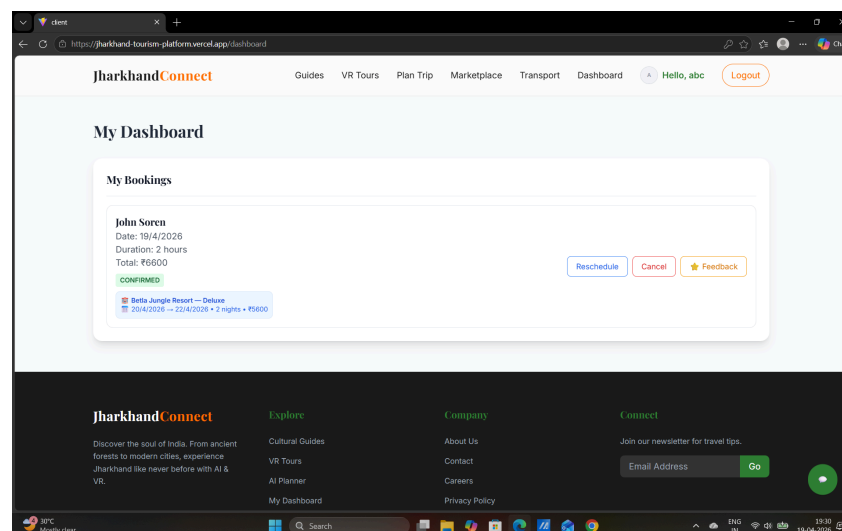


Figure 11.4: User and Admin Dashboard

11.5 Guide Booking

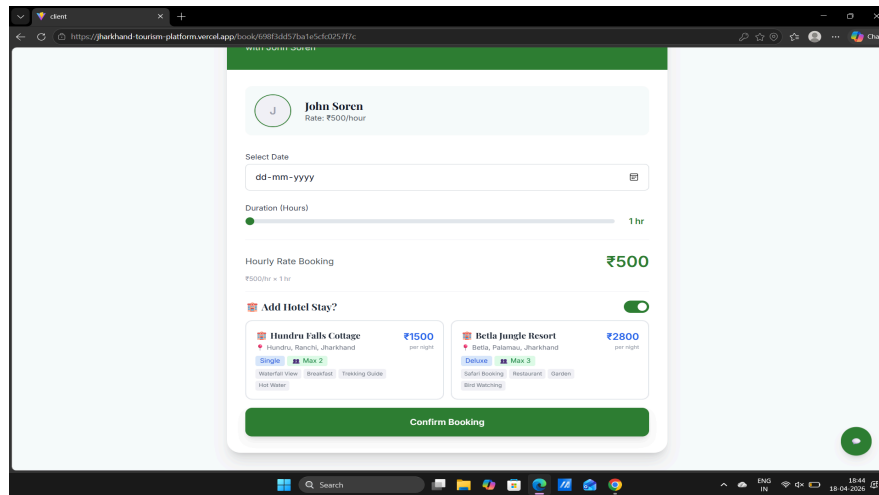


Figure 11.5: Guide Booking Page

11.6 VR Tours

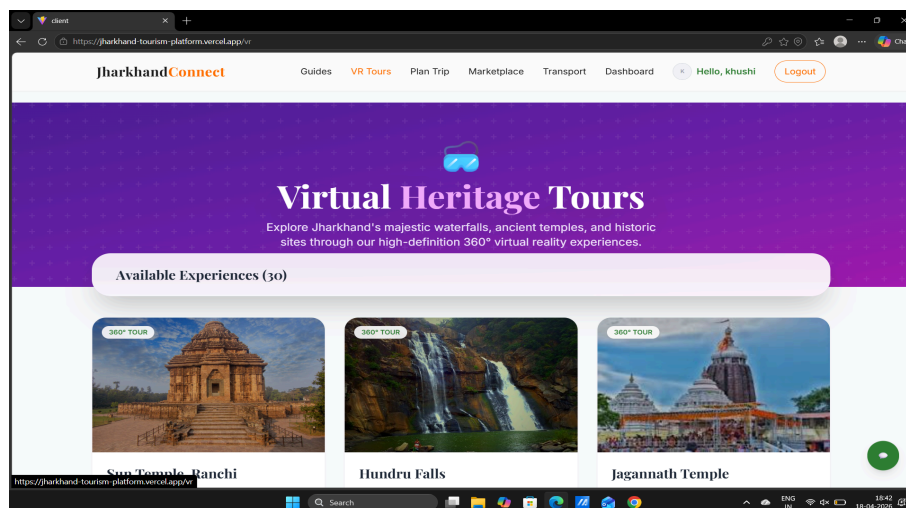


Figure 11.6: VR Heritage 360° Virtual Tour

11.7 Plan Trip

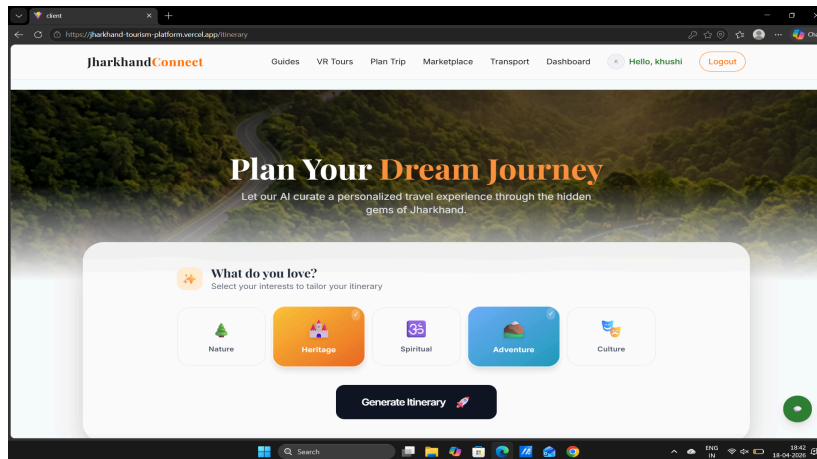


Figure 11.7: AI-Powered Trip Planner

11.8 Marketplace

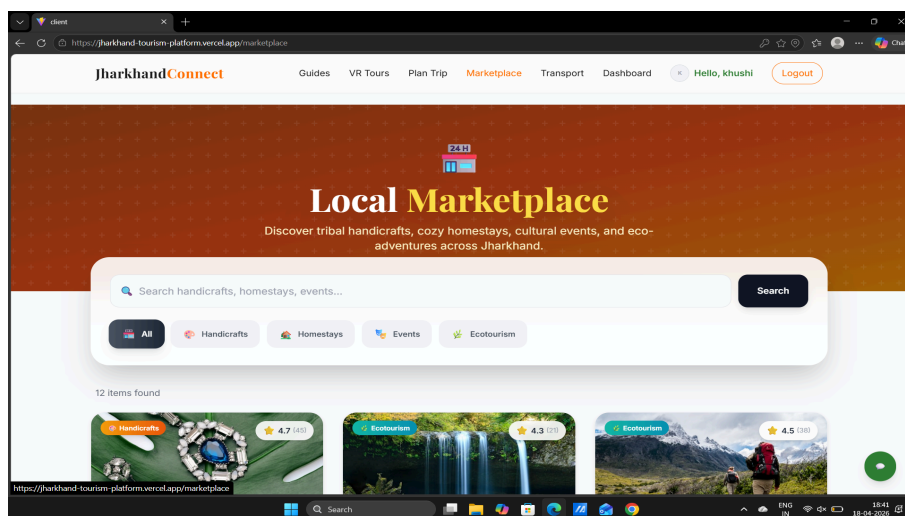


Figure 11.8: Local Handicraft Marketplace

11.9 Transport Booking

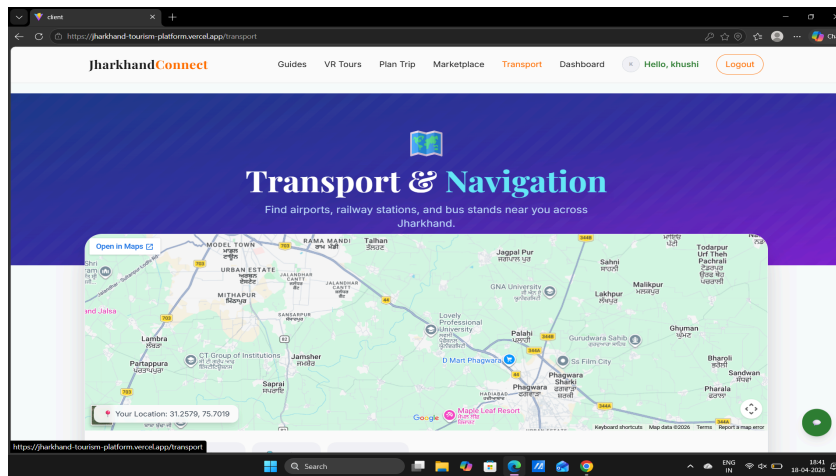


Figure 11.9: Transport Booking Page

11.10 AI Chatbot

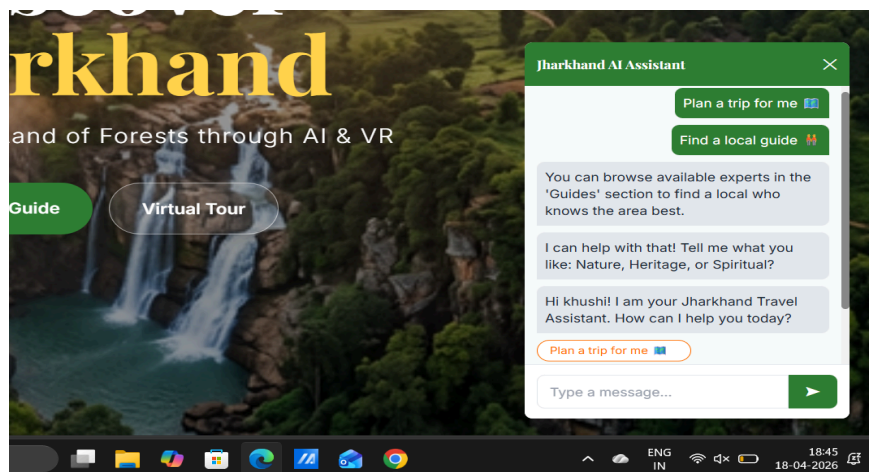


Figure 11.10: AI Multilingual Chatbot

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